

Project Planning

OptiCrop — Smart Agricultural Production Optimization Engine

Approach

Given a solo, 7-day timeline, the project was planned as a single sequential sprint rather than parallel team workstreams — each day builds directly on the previous day's output, with no step started before its dependency is verified working.

Day-by-Day Plan

Day	Focus	Key Deliverable
Day 1	Environment setup, repo + dataset	Working Python/Flask environment, dataset loaded
Day 2	Exploratory Data Analysis	Univariate, bivariate, multivariate analysis complete
Day 3	Data pre-processing	Null check, outlier handling, train/test split
Day 4	Model building	K-Means clustering + 4 classifiers trained and compared
Day 5	Model finalization	Best model (Random Forest) saved as .pkl artifacts
Day 6	Flask application	3-page web app (Home, About, Find Your Crop) working end-to-end
Day 7	Testing, documentation, submission	GitHub repo finalized, screenshots, demo recorded